

INHA UNIVERSITY IN TASHKENT
School of Computers and Information Engineering
Fall semester 2023
Assignment # 3

ID U2210251
Name Yavohirbek Xatamov
Group CSE-22-08

Data Structures (SOC2010)

Submission Deadline: 06 Oct 2023 (23:00 Hr.)

Problems based on Stack

NOTE: [Submissions not conforming to the given format will not be evaluated]

1. You are required to **take print out of this sheet** and solve (**handwritten**) the questions on sheet.
2. **Copying** of solution is strictly prohibited. In case it is found, then zero marks will be awarded.
3. Late submissions are not accepted.

Questions

Ques 1: Consider the following stack, where STACK is allocated N=6 memory cells

STACK: AAA, DDD, EEE, FFF, GGG.....

Describe the stack as the following operations take place:

	Operation	STACK	Comment (if any)
		AAA, DDD, EEE, FFF, GGG top	
1. PUSH(KKK)	PUSH(KKK)	KKK, AAA, DDD, EEE, FFF, GGG top	
2. POP()	POP()	AAA, DDD, EEE, FFF, GGG top	
3. PUSH(LLL)	PUSH(LLL)	LLL, AAA, DDD, EEE, FFF, GGG top	
4. PUSH(SSS)	PUSH(SSS)	SSS, LLL, AAA, DDD, EEE, FFF, GGG top	
5. POP()	POP()	LLL, AAA, DDD, EEE, FFF, GGG top	
6. PUSH(TTT)	PUSH(TTT)	TTT, LLL, AAA, DDD, EEE, FFF, GGG top	overflow

Ques2: Convert the following INFIX expressions to their PREFIX equivalents Show the step by step execution and status of STACK

Example (A * B) + (C / D) - (D + E)			(A - B) + D / ((E + F) * G)		
Write here method used (Method 1 or Method 2)					
Char	Stack	Expression	Char	Stack	Expression
	)			)	
)	)		)	)	
E	)	E	G	)	G
+	)	E	*	)	G
D	)	ED	)	)	G
/	)	ED/	F	)	GF
+	)	ED/ +	+	)	GF
)	)	ED/ +	E	)	GFE
D	)	ED/ + D	(	)	GFE +
/	)	ED/ + D	(	)	GFE + *
C	)	ED/ + D C	/	)	GFE + */
-	)	ED/ + D C -	D	)	GFE + */ D

+	) - +	ED + DCI	+	) +	GFE + *DI
)	) - +)	ED + DCI	)	) +)	GFE + *DI
B	) - +)	ED + DCIB	B	) +)	GFE + *DI/B
*	) - +)*	ED + DCIB	-	) +)-	GFE + *DI/B
A	) - +)*	ED + DCIBA	A	) +)-	GFE + *DI/BA
C	) - +	ED + DCIBAX	C	) +	GFE + *DI/BA -
C		ED + DCIBAX + -	C		GFE + *DI/BA - +
- + * ABCD + DE			+ - AB/DX + EFG		

(A - 2 * (B + C) / D * E) + F			14 / 7 * 3 - 4 + 9 / 2		
Char	Stack	Expression	Char	Stack	Expression
	)			)	
F	)	F	2	)	2
+	) +	F	1	) /	2
)	) +)	F	8	) /	28
E		FE	+	) +	28 /
*	) +)*	FE	4	) +	28 / 4
D	) +)*	FED	-	) + -	28 / 4
1	) +)* /	FED	3	) + -	28 / 4 3
)	) +)* /)	FED	*	) + - *	28 / 4 3 *
C	) +)* /)	FEDC	7	) + - *	28 / 4 3 *
+	) +)* /) +	FEDC	1	) + - * /	28 / 4 3 *
B	) +)* /) +	FEDCB	14	) + - * /	28 / 4 3 * 14
C	) +)* /	FEDCB +			28 / 4 3 * 14 / * - +
*	) +)* / *	FEDCB +			+ - * / 14 * 34 / 92
2	) +)* / *	FEDCB + 2			
-	) +)* -	FEDCB + 2 * /			
A	) +)* -	FEDCB + 2 * / A			
C	) +	FEDCB + 2 * / * A -			
		+ - A * / * 2 + BCDEF			

Ques 3: Evaluate the PREFIX expressions found in question 2: (show step by step using STACK)
[assume A=2, B=3, C=2, D=4, E=3, F=2, G=3]

[AB*CD/+DE+ -] → 23*24/+43+ -			- + * ABCD + DE → - + * 23/24 + 43		
Char	Stack	Operation	Char	Stack	operation
2	2		3	3	
3	2, 3		4	3, 4	
*	6		+	7	4 + 3
2	6, 2		4	7, 4	
4	6, 2, 4		2	7, 4, 2	
1	6, 2	4 / 2	1	7, 0.5	2 / 4

+	8	2+6	3	7, 0.5, 3	
4	8, 4		2	7, 0.5, 3, 2	
3	8, 4, 3		*	7, 0.5, 6	2 * 3
+	8, 7	3+4	+	7, 6.5	6+0.5
-	-1	4-8	-	-0.5	7-6.5

+ - A B / D * + E F G → + - 2 3 / 4 * + 3 2 3			+ - A * / * 2 + B C D E F → + - 2 * / * 2 + 3 2 4 3 2		
Char	Stack	operation	Char	Stack	operation
3	3		2	2	
2	3, 2		3	2, 3	
3	3, 2, 3		4	2, 3, 4	
+	3, 5	3+2	2	2, 3, 4, 2	
*	15	5*3	3	2, 3, 4, 2, 3	
4	15, 4		+	2, 3, 4, 5	3+2
/	0.3	4/15	2	2, 3, 4, 5, 2	
3	0.3, 3		*	2, 3, 4, 10	2*5
2	0.3, 3, 2		/	2, 3, 2.5	10/4
-	0.3, -1	2-3	*	2, 7.5	2.5*3
+	-0.7	-1+0.3	2	2, 7.5, 2	
			-	2, -5.5	2-7.5
			+	-3.5	-5.5+2

Ques 4: Write a pseudo code to copy the contents of first stack "A" in the second stack "B". (Note that the order of elements must be preserved)

CODE

```

tempStack = []
while not A.isEmpty();
    element = A.pop();
    tempStack.push(element);
while not tempStack.isEmpty();
    element = tempStack.pop();
    B.push(element)
return B.

```

#comment(if any).